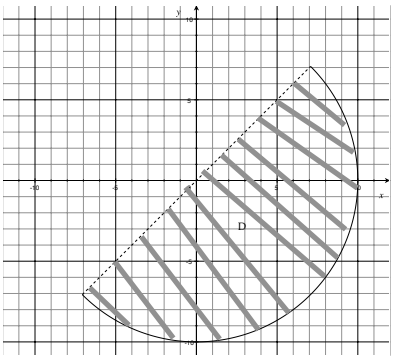
Exam II Answers
Math 126 C Winter 2019

1. $D = \{(x, y) : x^2 + y^2 \leq 10 \text{ and } x > y\}$



2. $\dfrac{\partial z}{\partial x} = \dfrac{2xz\cos(x^2z)}{8ye^z - x^2\cos(x^2z)}$

   $\dfrac{\partial z}{\partial y} = \dfrac{8e^z}{x^2\cos(x^2z) - 8ye^z}$

3. There are saddle points at $\left(0, -\sqrt{\frac{1}{2}}\right)$ and $\left(0, \sqrt{\frac{1}{2}}\right)$, a local max at $(1, 0)$, and a local min at $(-1, 0)$.

4. $\displaystyle\int_1^2 \int_1^{x^2} \left(xe^y + \frac{1}{x^3}\right) dy\, dx = \frac{1}{2}e^4 + \ln(2) - 2e - \frac{3}{8}$

5. $\displaystyle V = \int_0^3 \int_{\frac{x}{3}}^{3-\frac{2}{3}x} xy\, dy\, dx$